PROJECT DOCUMENTATION

# Amrit Dairy Farms

## Sales & Inventory Dashboard in Excel

Introduction, dataset description, data preparation, visualizations, insights and conclusions for the Amrit Dairy Farms Excel dashboard (AmritDairy_dashboard.xlsm).

**Data period:** 1 Jan 2019 – 28 Dec 2022 **Records:** 4,325 **Tool:** Microsoft Excel (Power Query, PivotTables, Slicers, VBA)

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# 1. Introduction

## 1.1 Background

Amrit Dairy Farms sells ten dairy products (milk, curd, paneer, ghee and others) through retail, wholesale and online channels to customers in 15 Indian states and union territories. Sales, stock and reorder information was held in a single flat file, which made it hard to answer everyday questions about what sells, where, and what needs restocking.

## 1.2 Objectives

- Prepare the raw sales and inventory data for analysis in Excel.
- Build an interactive one-page dashboard covering revenue, quantity sold, sales channels, geography, trend over time and stock health.
- Let users filter by brand, year and quarter, and reset the view with one click.
- Summarise the key findings and recommend actions.

## 1.3 Tools and approach

The CSV was loaded into Excel with Power Query and stored as an Excel Table. Helper columns (Month, Year, Quarter, Reorder Flag) were added, thirteen PivotTables were built on a Pivots sheet, and charts and KPI cards were assembled on a Dashboard sheet. Year and Quarter slicers control the pivots, a brand drop-down with Submit and Reset buttons (VBA macros) drives a Filtered Data sheet, and the workbook is saved as .xlsm so the macros work.

## 1.4 Workbook structure

| Sheet | Purpose |
|---|---|
| dairy_dataset | Cleaned data table (4,325 rows) plus helper columns and the brand list used by the drop-down. |
| Filtered Data | Rows for the brand selected on the Dashboard, retrieved with VLOOKUP formulas. |
| Pivots | Thirteen PivotTables that feed every chart and KPI card. |
| Dashboard | The presentation sheet: KPI cards, six charts, Year and Quarter slicers, brand drop-down, Submit and Reset buttons. |

# 2. Dataset Description

The source file (Amrit_dataset.csv) has 4,325 rows and 23 columns. Each row is one batch of a product, combining production, sales and stock information. Dates run from 1 January 2019 to 28 December 2022. Quantities are in litres (liquid products) or kilograms (solid products).

## 2.1 Columns

| Column | Type | Description |
|---|---|---|
| Location | Text | State where the farm is located (15 values). |
| Total Land Area (acres) | Number | Farm land area, 10 to 1,000 acres. |
| Number of Cows | Integer | Herd size, 10 to 100. |
| Farm Size | Text | Small, Medium or Large. |
| Date | Date | Transaction date. |
| Product ID / Product Name | Int / Text | Ten products, one ID each (1 Milk … 10 Ghee). |
| Brand | Text | Eleven brands, including Amul, Mother Dairy, Raj and Sudha. |
| Quantity (liters/kg) | Number | Quantity produced in the batch. |
| Price per Unit | Number | List price per litre or kg (₹10 to ₹100). |
| Total Value | Number | Quantity × Price per Unit. |
| Shelf Life (days) | Integer | Days from production to expiry (1 to 150). |
| Storage Condition | Text | Refrigerated, Frozen, Ambient, Tetra Pack or Polythene Packet. |
| Production Date / Expiration Date | Date | Batch production and expiry dates. |
| Quantity Sold (liters/kg) | Integer | Units sold in the transaction. |
| Price per Unit (sold) | Number | Actual selling price per unit. |
| Approx. Total Revenue(INR) | Number | Quantity Sold × Price per Unit (sold). |
| Customer Location | Text | State of the buyer (15 values). |
| Sales Channel | Text | Retail, Wholesale or Online. |
| Quantity in Stock (liters/kg) | Integer | Stock on hand after the sale. |
| Minimum Stock Threshold (liters/kg) | Number | Stock level that should trigger a reorder. |
| Reorder Quantity (liters/kg) | Number | Suggested quantity to reorder. |

## 2.2 Overall figures (all brands)

| Measure | Value | Note |
|---|---|---|
| Total revenue | ₹58,734,648 | Sum of Approx. Total Revenue |
| Units sold | 1,073,011 | Litres and kg combined |
| Average revenue per record | ₹13,580 | |
| Total reorder quantity | 471,892 | All records |
| Products / brands / locations | 10 / 11 / 15 | |

# 3. Data Cleaning and Preparation

## 3.1 Quality checks

Before building anything, the data was tested for completeness and internal consistency. Most checks passed; three issues need to be kept in mind when reading the results.

| Check | Result | Comment |
|---|---|---|
| Missing values | None | All 4,325 × 23 cells are filled. |
| Duplicate rows | None | No exact duplicates. |
| Total Value = Quantity × Price | Passes | Maximum difference is below ₹0.000001. |
| Revenue = Quantity Sold × Price (sold) | Passes | Same. |
| Expiration − Production = Shelf Life | Passes | 100% of rows. |
| Quantity Sold ≤ Quantity produced | Passes | No row sells more than it produced. |
| Product ID ↔ Product Name | Passes | One-to-one mapping. |
| Sale date after expiry date | Flag | 2,564 rows (59.3%) are dated after the expiry date. |
| Zero stock | Flag | 18 rows have no stock left. |
| Farm attributes per Location | Flag | Cow count varies within a location (81 to 91 different values), and Farm Size is unrelated to land area. |

## 3.2 Preparation steps in Excel

- **Loaded with Power Query** and stored as an Excel Table (dairy_dataset), so every formula and pivot refreshes when data changes.
- **Data types set:** dates as Date, quantities and prices as Number, categories as Text.
- **Added Month, Year and Quarter:** Month = TEXT(Date,"mmm"), Year = YEAR(Date), Quarter = QTR-1 to QTR-4 from the month. These feed the monthly trend and the Quarter slicer.
- **Added Reorder Flag:** "Reorder" when Quantity in Stock is below the Minimum Stock Threshold, otherwise "OK".
- **Added a Helper key** (brand name plus a running count) so the Filtered Data sheet can pull every row for the selected brand with VLOOKUP.
- **Built a brand list** on the data sheet to feed the Dashboard drop-down.
- **Created thirteen PivotTables** on the Pivots sheet from one shared pivot cache, and connected the Year and Quarter slicers to them.

## 3.3 Treatment of the flagged issues

- **Sales after expiry:** kept, because revenue and quantity are internally consistent. The pattern (almost 6 in 10 rows) is more likely a data-generation artefact than real late sales, so expiry-based analysis was not included on the dashboard.
- **Zero stock:** kept; these are genuine stock-outs and are counted as below threshold.

- **Farm attributes:** not used. Because cow count and farm size are not stable within a location, revenue per cow or per farm size would be misleading.

**Note on the data:** values such as quantities, prices and land area are spread almost evenly across their ranges, which suggests the dataset is sample or synthetic data. The findings below describe this dataset and should be validated before real business decisions.

# 4. Dashboard and Visualizations

## 4.1 Dashboard layout

The Dashboard sheet is one page. KPI cards across the top show Revenue, Quantity and Total Reorder Quantity. Below are charts for monthly revenue, revenue by product, channel share, product revenue by channel, stock versus threshold and top five states. Year and Quarter slicers sit on the right, together with a brand drop-down and Submit button; Reset returns the view to its default.

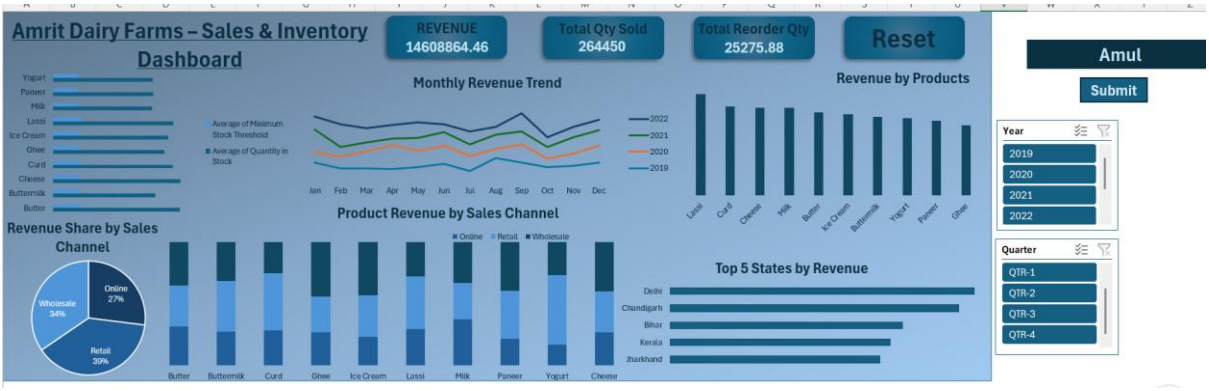



*Figure 1. The finished dashboard with Amul selected. The Quantity card in this capture shows the all-brand total; the saved workbook shows 264,450 for Amul.*

**How the brand view works:** the brand chosen in the drop-down is applied when Submit is pressed. In the saved file Amul is selected, so the KPI cards read ₹14,608,864 revenue, 264,450 units and 113,983 total reorder quantity, which are Amul's figures, not the company total of ₹58.7M. The Year and Quarter slicers filter on top of the brand view.

## 4.2 Visualizations and what they answer

| Visual | Chart type | Question it answers |
|---|---|---|
| KPI cards | Linked text cards | How large is the business in the selected view (revenue, units, reorder quantity)? |
| Monthly Revenue Trend | Line, one line per year | Is there seasonality, and is revenue growing year on year? |
| Revenue by Products | Column | Which products earn the most? |
| Revenue Share by Sales Channel | Pie | How is revenue split across Retail, Wholesale and Online? |
| Product Revenue by Sales Channel | 100% stacked column | Which channel is strongest for each product? |
| Top 5 States by Revenue | Bar | Which markets matter most? |
| Stock vs Minimum Threshold | Clustered bar | Which products hold too little or too much stock? |

| Visual | Chart type | Question it answers |
|---|---|---|
| Year and Quarter slicers | Slicers | Narrow every pivot and chart to a period. |

## 4.3 Supporting charts (full dataset, all brands)

The charts below are drawn from the complete dataset so the findings in the next section can be seen at company level. They show the same measures as the dashboard.

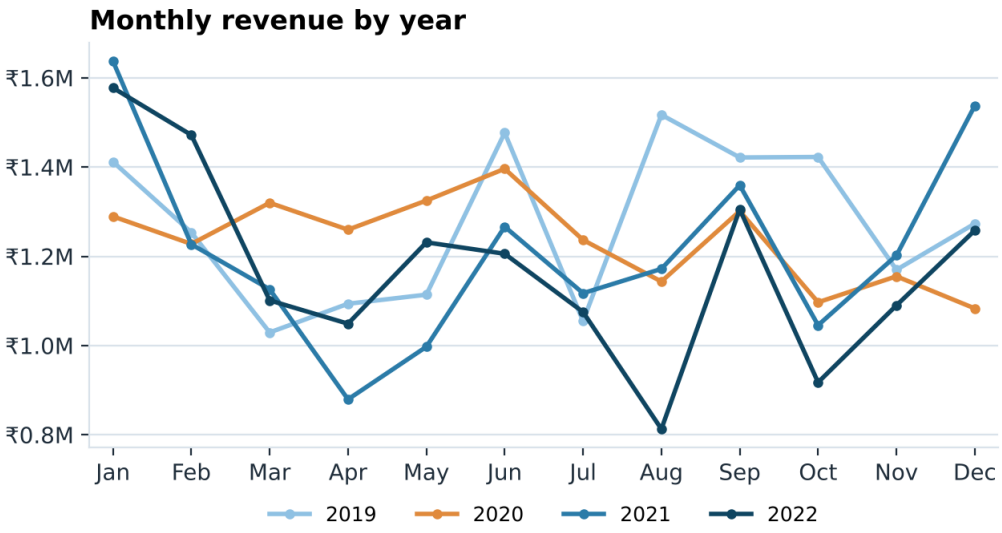



*Figure 2. Monthly revenue by year, all brands.*

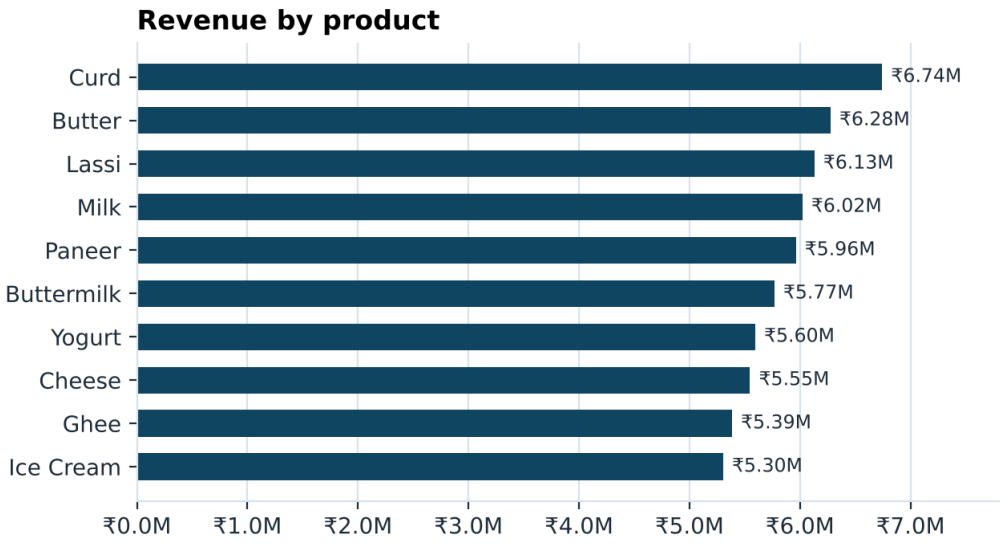



*Figure 3. Revenue by product.*

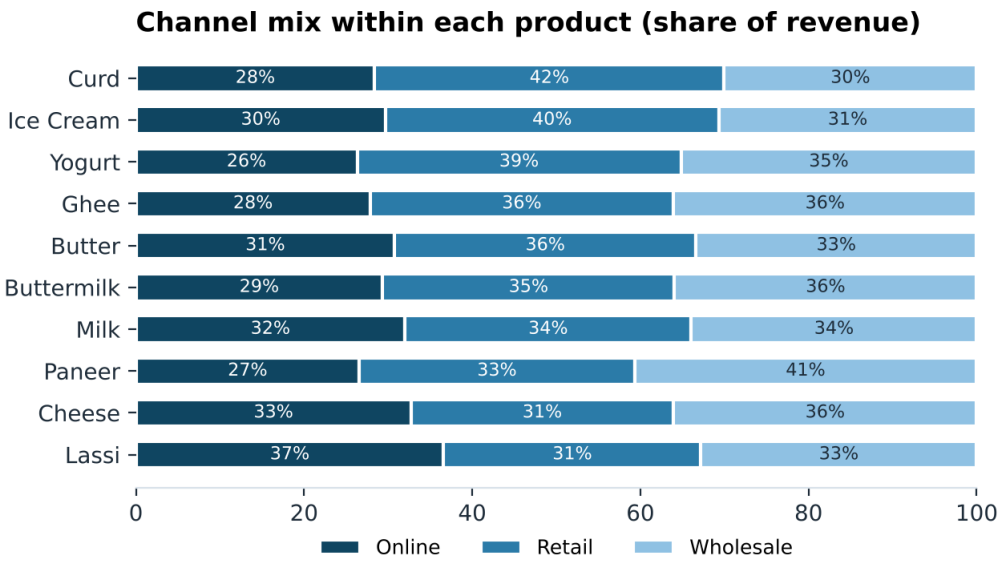



*Figure 4. Share of each product's revenue by sales channel.*

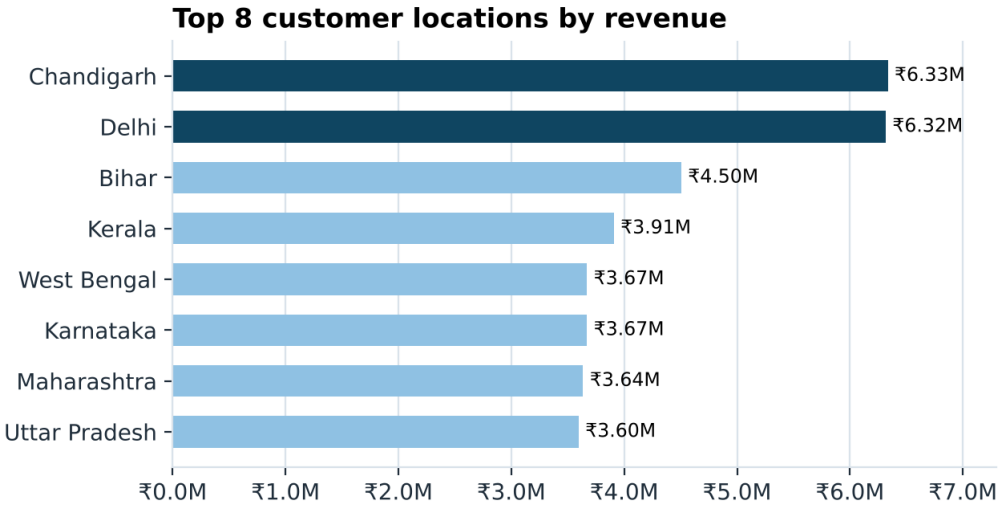



*Figure 5. Top 8 customer locations by revenue.*

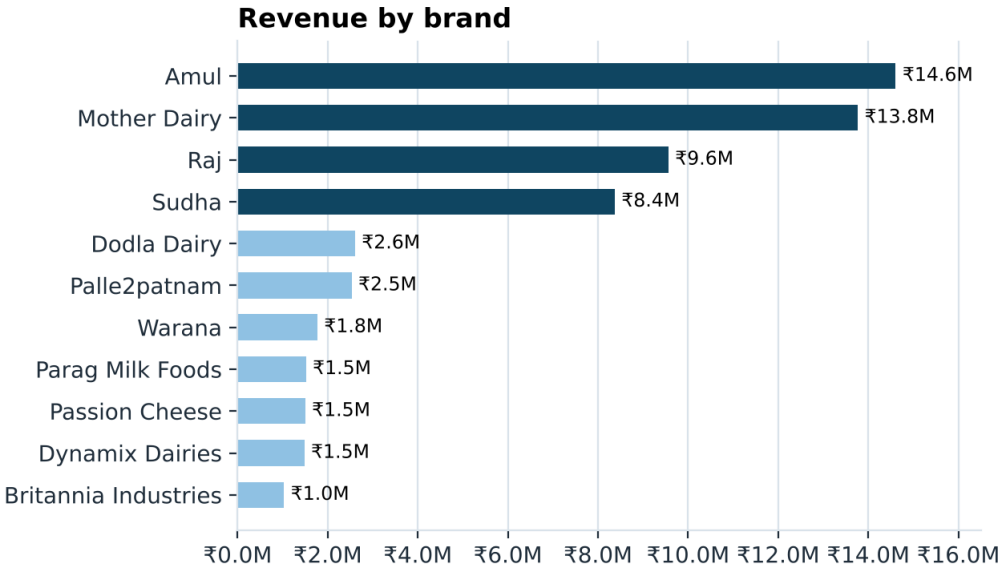



*Figure 6. Revenue by brand.*

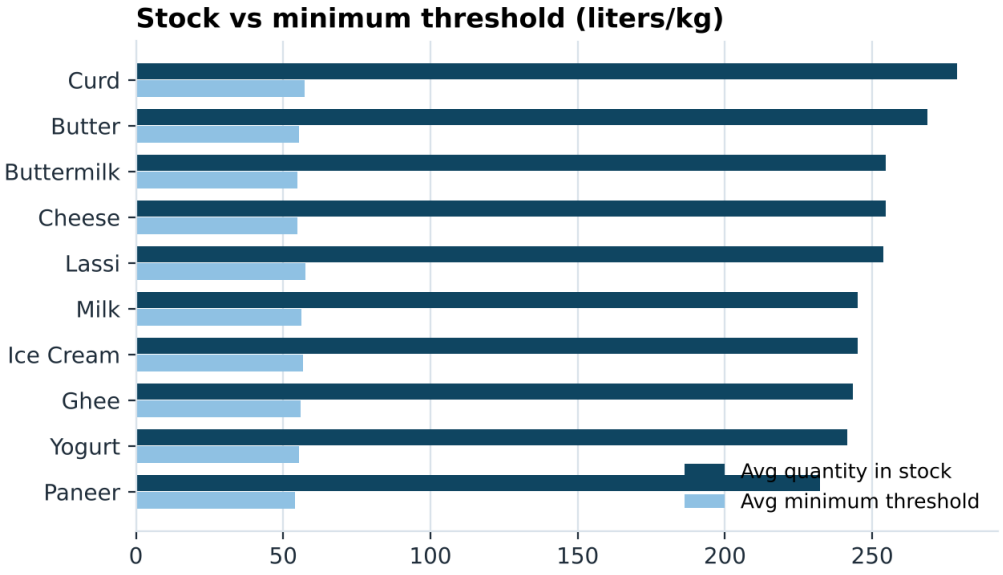



*Figure 7. Average stock against minimum threshold, by product.*

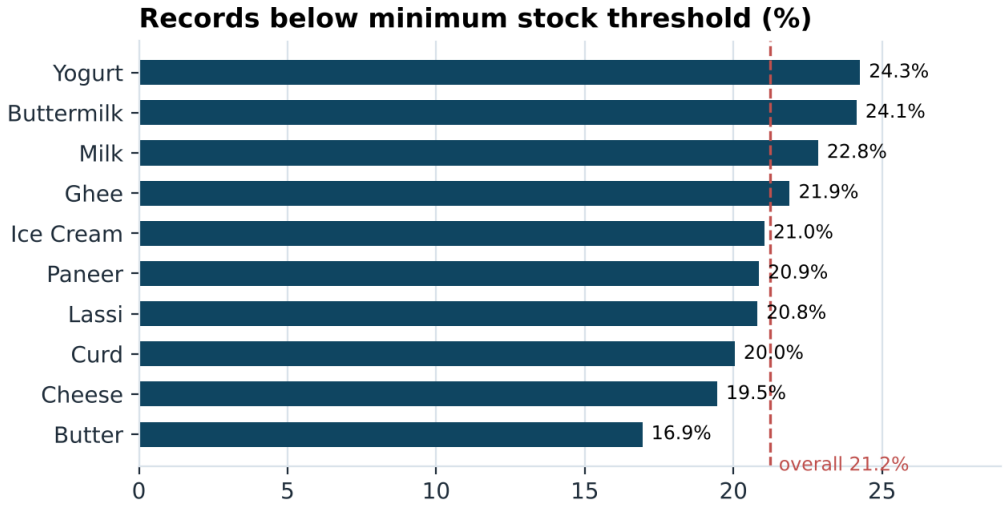



*Figure 8. Share of records that fall below the minimum stock threshold.*

# 5. Insights

Unless stated otherwise, figures cover all brands and all four years (4,325 records, ₹58.73M revenue).

## 5.1 Revenue and products

- **Revenue is spread evenly across products.** Curd leads with ₹6.74M (11.5%) and Ice Cream is lowest at ₹5.30M (9.0%), a gap of only 2.5 percentage points. Butter (10.7%), Lassi (10.4%) and Milk (10.3%) follow Curd.
- **Average record value is steady.** It ranges from ₹12,541 (Ice Cream) to ₹14,562 (Butter) around the overall ₹13,580.

## 5.2 Trend and seasonality

- **Revenue has fallen every year.** Annual revenue went from ₹15.24M (2019) to ₹14.84M, ₹14.57M and ₹14.10M (2022), a drop of 7.5% over three years (−2.7%, −1.8% and −3.2% year on year). Records fell from 1,109 to 1,035, while the average record value stayed near ₹13,700, so the decline comes from fewer transactions rather than lower prices.
- **January is the strongest month** (₹5.92M), followed by September (₹5.39M) and June (₹5.35M). April is the weakest (₹4.28M). Q1 delivers 26.7% of revenue against about 24–25% for each other quarter.

## 5.3 Channels

- Retail (35.5%) and Wholesale (34.4%) are almost level; Online is smallest at 30.1%, with the lowest average record value (₹12,879 against ₹14,116 for Retail).
- Channel strengths differ by product: Curd sells most through Retail (₹2.80M), Paneer through Wholesale (₹2.43M) and Lassi through Online (₹2.24M, its largest channel).

### 5.4 Geography

- **Delhi and Chandigarh together bring in 21.6% of revenue** (₹6.32M and ₹6.33M by customer location, about 10.8% each). The other 13 locations each contribute 5.0% to 7.7%.
- Bihar is third (₹4.50M) with the highest average record value of any location (₹15,863), suggesting larger orders rather than more orders.

### 5.5 Brands

- Amul (24.9%, ₹14.61M) and Mother Dairy (23.4%, ₹13.77M) lead; with Raj (16.3%) and Sudha (14.3%) the top four brands hold 78.9% of revenue. The remaining seven brands share 21.1%.
- Warana has the highest average record value (₹17,014) but only 104 records; Britannia Industries has the lowest (₹11,699).

### 5.6 Inventory

- **21.2% of records (918) sit below their minimum threshold,** and 18 have no stock. Yogurt (24.3%) and Buttermilk (24.1%) are worst; Butter is best at 16.9%.
- On average, stock (252 units) is well above the threshold (56 units), yet one record in five is below it. Stock levels are uneven rather than uniformly low.
- Only about half of what is produced is sold (sell-through 49.6%), and it is similar for every product (47% to 51%).
- Reorder quantity totals 471,892 across all records. Restricting it to the 918 records flagged "Reorder" gives 98,041, which is a more actionable figure than the total on the dashboard card.

### 5.7 Pricing and storage

- Selling price is on average within 0.1% of list price, so there is no visible discounting or price premium in the data.
- Refrigerated products produce 57.2% of revenue, Frozen 23.4%, Ambient (Ghee only) 9.2%, and Polythene Packet and Tetra Pack (both Milk) about 5% each.

## 6. Conclusions

Amrit Dairy Farms has a balanced portfolio: no product, channel or state other than Delhi and Chandigarh dominates, and four brands generate nearly four-fifths of revenue. The main concerns are a revenue decline in each of the last three years, driven by fewer transactions, and inventory that is uneven, with one in five records below threshold while about half of production goes unsold.

The Excel dashboard brings these views together in one page with brand, year and quarter filtering, so a manager can move from company totals to a single brand or season in a few clicks.